

NSF AI Institute for ARtificial and Natural Intelligence ARNI



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Response to the NSF Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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Introduction

NSF ARNI's mission is to reach the next level of brain-like AI. We aim to achieve strong generalization by connecting the remarkable progress in AI systems with the revolution that is happening in our understanding of the brain. This interdisciplinary research is what we call "NeuroAI." Progress in this field has the potential to transformatively impact the economy and society in the next decade.

Despite stunning advances, there are three huge weaknesses for AI systems: they depend on enormous training data, burn huge amounts of energy, and, most critically, can make egregious errors. We seek to overcome these obstacles by drawing inspiration from the brain's architectures, circuits, and learning mechanisms. Our goal is to develop AI with stronger generalization and more human-like efficiency and adaptability.

The current boom in AI in fact is driven by neural network models inspired by biological neural networks. The idea was that by taking many artificial, simplified nerve cells, or neurons, and learning how to connect them, we could do hard computations like the human brain. This learning-based approach to AI was totally different from earlier approaches based on logic and design. Eventually, with enough data and computational power, the neural network models started working. Once that idea took root, massive engineering efforts eventually led to successes in language models, generative images and video, and robotics. This approach was so successful compared to other approaches that one of the luminaries of the field and recent Turing Award winner, Richard Sutton, penned a short essay called "the Bitter Lesson" in which he said that our cleverest ideas didn't help: only more data and bigger, brute force compute mattered. We had to "go big." This current approach has its costs.

First, progress was powered by voracious data centers, burning city-scale energy and using thousands of times more energy than our brains. Some engineering efforts, like DeepSeek from China, have claimed to dial back the amount of energy required, but the consumption of energy remains huge.

Second, even with unlimited energy, we're running out of data. Language models are now trained on essentially all text ever published by humans. We cannot just use machines to generate more text without poisoning the data by snowballing mistakes.

Third, our best AI systems remain largely inscrutable, still making errors that range from amusing to appalling. They cannot be trusted with important decisions in medicine, economics, business, law, and many other domains where common sense and reasoning are critical.

We believe NeuroAI is poised to address many of these challenges. AI was originally revolutionized by brain-inspired neural networks. We now have much richer knowledge of the brain's inner workings. The time is right for a NeuroAI revolution, one that incorporates our impressive new neuroscience knowledge about biological intelligences honed over eons.

In the last decade, neuroscience has made groundbreaking progress in measuring and understanding neural activity and learning. We can now measure up to a million neurons at the

same time in a functioning brain. We have now reconstructed the wiring of hundreds of thousands of neurons. With these data, our analyses have discovered core principles of how countless neurons act together to encode information about the world, make decisions, and drive complex movements. Neurotechnology continues to improve rapidly.

However, these neuroscience insights have yet to be integrated into AI. Through biologically inspired architectures, neuromorphic computing, and principles of cortical learning, NeuroAI research can lead to breakthroughs in continual learning, energy-efficient inference, and explainable AI. While the impressive and attention-grabbing models of the last couple years made advances through heavy compute and data, our aim is to “Go Small”, to take the brain as an existence proof that we can use to inspire advances in these directions.

Additionally, Large Language Models (LLMs) and Vision Language Models (VLMs) have created a paradigm shift in AI, taking systems to a level far beyond what was possible even five years ago. These models, however, still have significant problems in performance. They tend to hallucinate, making up facts out of thin air, causing difficulties in using them in deployment; for example, recently Apple was forced to take down its news summarizer after it incorrectly covered a BBC news report. Models operate as black boxes making explainability and interpretability difficult. They encode the distribution of their training data, making accurate generalization difficult when the distribution shifts.

There is clear competition internationally, including European initiatives like the Human Brain Project and China’s aggressive state-backed investments in neuromorphic chips, brain-inspired algorithms, and next-generation AI systems. Leadership in the face of such unconventional rivals calls for stronger interdisciplinary collaborations between AI, neuroscience, and cognitive science — as fostered by our NSF AI Institute.

This document presents our recommendations for incorporating NeuroAI into the NSF AI Action Plan.

Our thesis:

Investing in interdisciplinary NeuroAI research at the intersection of AI, neuroscience, and cognitive science is critical for the U.S. to maintain its leadership in artificial intelligence, particularly as Europe and China make significant advances in this field.

Our Recommendations on Research and Development Priorities

Our recommendations can be divided according to the essential focus on the bidirectional interaction between AI and neuroscience/cognitive science. We first present priorities that address how cognitive science and neuroscience can inspire AI, and then on AI’s contributions to advancing these fundamental areas of research. We then discuss additional recommendations considering how AI research can play key roles in society and internationally.

I. Gaining inspiration from neuroscience and cognitive science to advance

Brain-Inspired AI Architectures

- NSF should prioritize research in biologically plausible neural networks at both the hardware and software level. Relevant approaches to explore include spiking neural networks (SNNs), neuromorphic computing, networks with recurrence and feedback, networks with excitatory and inhibitory cells and their subtypes, and roles of dendrites in computation.
- Strategically support commercialization of neuromorphic chips, which remains limited.
- Encouraging collaboration between AI researchers and neuroscientists to develop AI models that leverage brain-like processing for improved learning and generalization.
- Supporting the development of hybrid architectures combining traditional deep learning and neuroscience-inspired models.

Developing Neuroscience Foundation Models

- Create multimodal foundation models capable of translating and predicting neural activity across different brain regions and behaviors.
- Leverage self-supervised learning on large-scale datasets to enable cross-subject generalization.
- Improve BCI adaptability and robustness with minimal training data and cross-species learning.
- Ensure accessibility by providing open-source models and APIs for broad research adoption.

Establishing Neuroscience Benchmarks

- Launch benchmarking challenges to assess model performance on diverse neural datasets.
- Focus on evaluating generalizability, robustness, and prediction accuracy across species and recording conditions.
- Utilize models for quality control, detecting artifacts, and refining neural datasets.

Neuroscience-Informed Learning Models

- Prioritize funding research into biologically inspired learning mechanisms such as local rules of synaptic plasticity, along with possible gating of learning by modulatory factors and by inhibition of top-down inputs; memory consolidation; and predictive coding.
- Exploring the role of sleep and replay mechanisms in AI model optimization and continual learning.
- Consider modules that focus on different forms of learning, inspired by the major divisions of vertebrate intelligence. These include a cerebellum, which is involved in

online learning from feedback about errors, and basal ganglia, a center for reward learning and for procedural learning. The vertebrate brain also includes a forebrain or telencephalon, which in mammals has primarily become the cerebral cortex, which does associative learning and builds complex world representations. There is a rich interplay between the three. Understanding the principles behind this modular design might benefit AI, and conversely, exploring AI built upon such modules might help reveal these principles.

- Another modular design feature to explore is the division within the cortex between episodic and factual learning (“knowing that”), which depends on the hippocampus and its interplay with the rest of cortex; and motor and procedural learning (“knowing how”), which depends on the interplay between cortex and basal ganglia.
- Promote energy-efficient AI algorithms inspired by the brain’s adaptive learning processes.

Neuro-Symbolic AI for Explainability

- Encouraging the development of hybrid Neuro-Symbolic AI systems that integrate neural network learning with symbolic reasoning, inspired by hierarchical processing in the brain.
- Investigating how cognitive neuroscience insights can improve AI’s ability to abstract, reason, and generalize across tasks.
- Supporting research in AI models that mimic hierarchical decision-making observed in the human brain.

Advancing AI through Neural and Cognitive Mechanisms

- State-of-the-art models hallucinate and inherit skews based on their training data, preventing strong, reliable generalization.
- While humans and other animals learn continuously, gaining abilities in new tasks and domains while not losing past skills, current models forget when tuned to new data.
- Incorporating memory mechanisms in current AI models is a promising approach.
- Prioritize funding on continual learning to develop new learning approaches based on cognitive memory mechanisms. Improved evaluation of models’ ability to learn over time.

Energy-Efficient AI Inspired by Biology

- Encouraging research on neural network architectures and objectives that yield more energy-efficient models.
- Encouraging research on neural network models that can flexibly trade off different costs, such as energy, errors, and inference time.
- Supporting theoretical work on the energy-efficiency of neural computations in biological and engineered systems.
- Supporting research into neuromorphic computing hardware inspired by the brain’s low-power processing.

- Aligning NSF's AI hardware initiatives with brain-like computing models to enable sustainable AI advancements.
- Supporting faster commercialization of neuromorphic computing advancements.

Neuroscience-Inspired AI Safety and Robustness

- Exploring how biological mechanisms such as uncertainty estimation, adaptation, and fault tolerance can improve AI safety.
- Support development of AI models that incorporate robustness and resilience similar to biological cognitive processes.
- Supporting research in adversarial robustness using principles from the brain's error correction mechanisms.

II. Using AI to Advance Neuroscience and Cognitive Science

AI for Understanding Brain Function

- Promoting AI-driven analysis of large-scale neural datasets to accelerate neuroscience discoveries, leading to new hypotheses and insights about brain function, and improved understanding of cognitive processing.
- Supporting research to develop AI models that replicate cognitive functions such as attention, perception, and memory.
- AI for improved neural signal and behavioral video processing

AI for Mental Health and Neurological Disorders

- Advancing AI applications for interpreting neural data can assist in early diagnosis and treatment of neurological conditions such as Alzheimer's, Parkinson's, and epilepsy.
- Developing AI-powered mental health tools that leverage brain-inspired modeling for personalized treatment approaches.
- Addressing challenges related to transparency, and ethical considerations in AI-driven healthcare solutions.

AI and Brain-Computer Interfaces (BCIs)

- Investing in AI-driven BCIs for medical and assistive technologies, including neuroprosthetics, sensory and motor prostheses, neural population decoding methods, and neurostimulation protocols.
- and adaptive rehabilitation systems.
- Ensuring ethical AI development in BCIs, addressing concerns regarding privacy and security.
- Supporting real-time adaptive AI algorithms for improved human-AI interaction via neural interfaces. This requires algorithms for using both input and output hardware, reading from and writing to the brain.

Understanding Animal Cognition through AI

- AI models trained on human cognition can help study non-human intelligence and cognitive evolution.

III. Additional Priorities

Enhancing AI Benchmarks, Evaluation and Methods for Vision and Language Tasks

- Research is needed into new benchmarks for challenging tasks, along with appropriate evaluation methods.
 - This will enable extension of LLMs and VLMs to new, creative tasks, where they have been shown to perform poorly.
 - Research has shown that human evaluation of LLM output is flawed, likely due to the fluency of the language, and new human evaluation protocols are needed.
 - At the same time, automatic metrics that can accurately judge novel tasks are also needed (e.g., has stance been conveyed when generating perspective summaries?).
- Promising directions for improvement in hallucination includes grounding generation in intermediate knowledge, detecting and addressing subjectivity in the output, and the use of interacting agents that can critique each other through approaches like red-teaming,
- As we move to new tasks involving creativity, promising directions include synthetic data generation when paired with good verifiers.
 - Inference time scaling and post training are incredibly powerful for tasks with verifiers.
 - Developing verifiers for creative tasks is still in its infancy.
- Research in cognitive modeling can help in developing new approaches that are better aligned with human behavior, helping to guide a model in where to focus attention in data to avoid encoding skewed distributions and to learn how to update knowledge appropriately over time.
- For all methods, we need new approaches that are computationally more efficient.
 - Research into text diffusion models is in early stages and shows promise, in some cases showing similar performance as LLMs at a fraction of the cost.

Societal Implications of NeuroAI

- Addressing policy concerns related to AI systems that mimic human cognition and decision-making processes.
- Establishing guidelines for the responsible use of NeuroAI in critical applications, including medical and defense-related domains.
- Encouraging public engagement and discussion on the societal impact of AI systems modeled after human brain functions.

International Competition and the Need for U.S. Leadership in NeuroAI

- China, UK, and Switzerland, followed by the European Union, Canada, and Japan are **heavily investing in NeuroAI research, recognizing its strategic importance for future AI advancements**. Areas of focus pertain neuromorphic hardware, computational models of cognition, and frontier areas of research e.g, brain-inspired deep learning, reinforcement learning, spiking neural networks and robotics to name a few. In particular, China has launched major national initiatives in brain-inspired computing, neuromorphic chips, and cognitive AI, integrating neuroscience with AI to enhance efficiency, intelligence, and adaptability.
- The Chinese government has established dedicated research centers, significantly increased funding, and fostered collaborations between industry (Huawei, Cambricon technologies, Baidu Brain) and academia (Peking University, Fudan University) to accelerate progress in this domain. Their major initiatives include China Brain Project and the New Generation AI Development Plan (2017-2030) that prioritize human brain-inspired AI research.
- To maintain U.S. leadership in AI, it is crucial for NSF and other federal agencies to
 - provide sustained strong funding and infrastructure support for research at the intersection of AI, neuroscience, and cognitive science
 - Support the ever-growing AI hardware sector (BCIs and neuromorphic chips)
 - Invest on academia-industry consortia
 - workforce development

Training the Next Generation of Global Leaders in Neuroscience and AI

- **AI & Neuroscience Curriculum:** Support the development of specialized curricula at the intersection of AI, neuroscience, and cognitive science. For instance, ARNI, in collaboration with Neuromatch Academy, has been co-developing a NeuroAI course that explores shared principles of intelligence between brains and machines.
- **Interdisciplinary Training:** Establish joint AI-Neuroscience Ph.D. programs, postdoctoral fellowships, and research institutes to foster cross-disciplinary expertise and drive innovation at the interface of AI and neuroscience.

Industry-Academia Collaboration:

- **Public-Private Partnerships:** Facilitate collaboration between academia, industry, and government to accelerate AI advancements and ensure real-world applications of NeuroAI research.
- **Data Sharing:** Encourage companies to share available data-sets through mutual collaboration. Companies contributing data can benefit from AI-driven insights and collaborative research
- **Strengthening industry-academia collaborations:** Faster commercialization of AI innovations from academia
- **Joint Academia-Industry Research Consortia:** Encourage joint research initiatives where academic institutions, industry partners, and startups co-develop AI models and applications inspired by neuroscience.

- **Government-Industry seed funding and compute resources** provided to accelerate startup launch from academia.
- **AI-Neuroscience Talent Pipeline:** Develop targeted funding programs, **internships**, and fellowship opportunities to train the next generation of AI and neuroscience researchers in both academia and industry.
- **AI Regulation and Ethics Frameworks:** Work closely with industry stakeholders to create responsible AI policies that balance innovation with ethical and societal considerations.

Infrastructure and Computing Needs

- **High-Performance Computing for Neuroscience:** Invest further in computing clusters for large-scale neural simulations and brain-inspired AI training. These are strategic assets in the AI economy era.
- **Long-Term Research Funding:** Provide sustained support for foundational AI and neuroscience research beyond short-term, application-driven funding cycles. Brain-inspired, energy-efficient research will provide the innovating country with a lasting advantage in optimizing and advancing AI applications.
- **Data Sharing and Accessibility:** Establish standardized, secure platforms for sharing neuroscience and AI datasets while ensuring ethical use and privacy protections.

The Role of the 27 NSF-Funded National AI Research Institutes

The National AI Research Institutes were established to drive long-term interdisciplinary AI research and reinforce U.S. academic leadership and global competitiveness in an AI-driven world. **We strongly recommend expanding their role as strategic assets** within a National AI Plan for Innovation.

Each institute, with its unique research focus, serves as a critical hub for government, academia, and industry collaboration. Strengthening support and funding will enable these institutes to accelerate AI innovation, drive real-world applications, and foster the growth of startups that can become the next Google! Additionally, they play a pivotal role in training the next-generation AI workforce in interdisciplinary AI research and by developing field-defining, specialized curricula. An example is the co-developed ARNI-Neuromatch NeuroAI course for graduate students and postdoctoral fellows.

Expand Funding: Enhance the National AI Research Institutes’ capacity to drive innovation, commercialization, and support workforce development.

Leverage Specialized Expertise: Develop targeted, high-impact interdisciplinary training programs to nurture the future global AI leaders.

Support Research Commercialization: Strengthen pathways for translating AI research into impactful technologies, applications, and startup ventures.

Conclusion

NeuroAI has the potential to revolutionize AI by making it more efficient, interpretable, and aligned with human cognition. NSF should actively support interdisciplinary research between AI and neuroscience communities, fostering innovation in brain-inspired AI models and their responsible application. Our recommendations highlight the key areas where NeuroAI can contribute to the AI Action Plan, ensuring that future AI systems are both advanced and ethically sound.

Academic research has long driven American innovation, from the birth of Silicon Valley to Boston's emergence as a global biotech hub to the deployment of AI and energy advancements. These contributions fuel economic growth, create jobs, and secure lasting advantages for the U.S. in an evolving global landscape. Increased funding towards academic research alongside a stronger partnership with industry, will be able to advance AI and sustain US dominance in the field.

We welcome the opportunity to discuss these recommendations further and contribute to shaping the NSF AI Action Plan.